Introduction to Security

OSTEP 53

Sam Ogden

DRAFT

This slide deck is still under active development. Expect changes before it is finalized.

Lecture Objectives

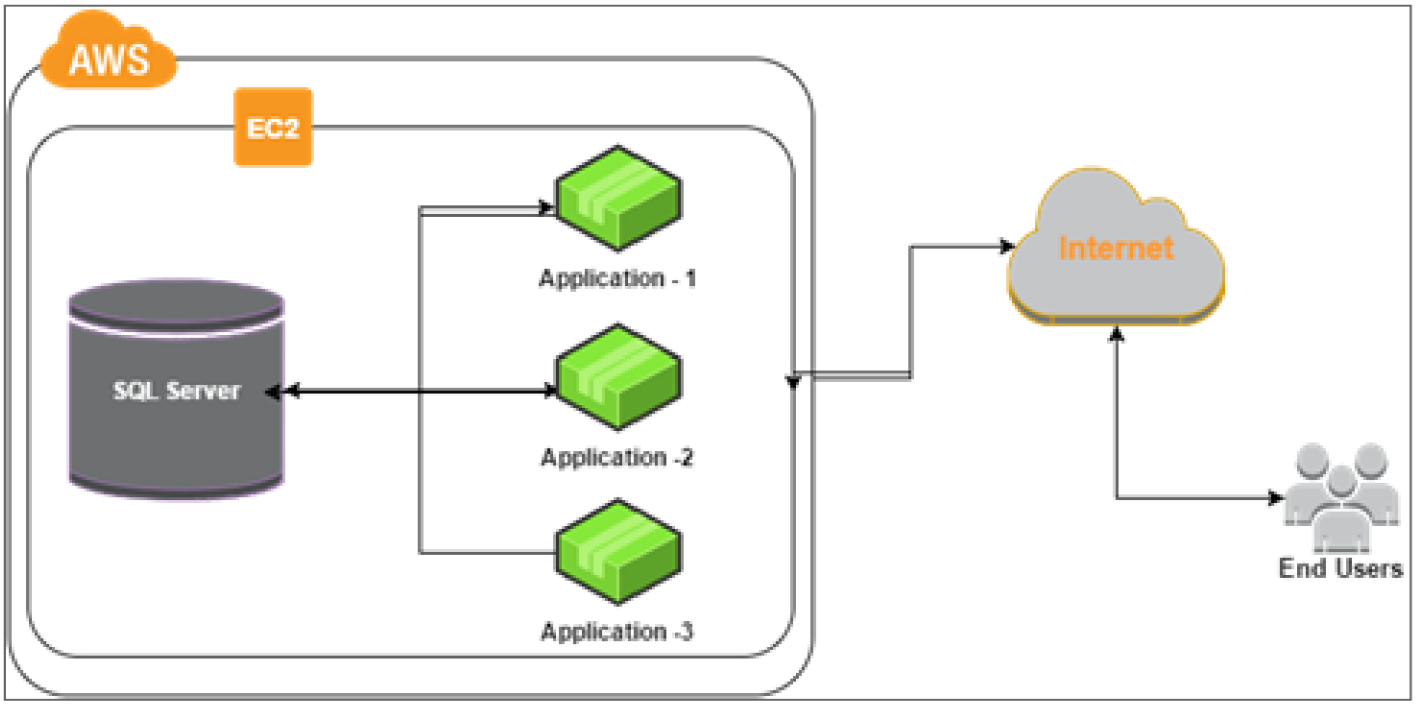
- **After this lecture, you should be able to:**
- **Explain why security is an essential consideration for operating systems**
- **Discuss the goals of security**
- **Core security concepts**

What we’ve done so far

- **So far we’ve virtualized resources:**
- Limited Direct Execution for processes
- Paging for memory
- **That keeps processes apart from each other**
- **But isolation alone does not answer how shared resources should be controlled**
- **Is that really enough?**

Resource Sharing is still important

- **Not all applications can share memory with just threads**
- **Sometimes we need to communicate between processes**



Many servers run databases in separate processes

What might we want to protect?

- **Process memory**
  - Passwords, personal information, execution pathways
- **Files**
  - User passwords are stored in a file. Who should be able to write?
- **I/O devices**
  - e.g. Who should be able to print?
  - e.g. Who should be able to send network messages?
- **Resource decisions!**
  - We need to make sure we are still the one allocating memory/compute

The OS already controls the resources we need to protect

That makes it the natural place to enforce security policy

Core Responsibilities

Because...

- **Owns memory management & sharing**
- **Starts/pauses/terminates processes**
- **Makes resources available**

Therefore...

- **Protect each process's memory**
- **Make reasonable choices**
- **Allocate them safely**

Security Goals

- Confidentiality: keep information secret
- Integrity: keep information correct
- Availability: keep the system up and running

Example: A cryptominer virus

- **Mines crypto using your GPU, without your permission**
    - Increases your power bill
    - Reduces your game’s FPS

What goals does this circumvent?
Availability

Security design is about tradeoffs

- The goal is not just to block attacks
- The goal is to protect the system without making it unusable
- Good security design reduces damage when something goes wrong

Design Principles

Keep it simple

- Economy of mechanism
- Open design

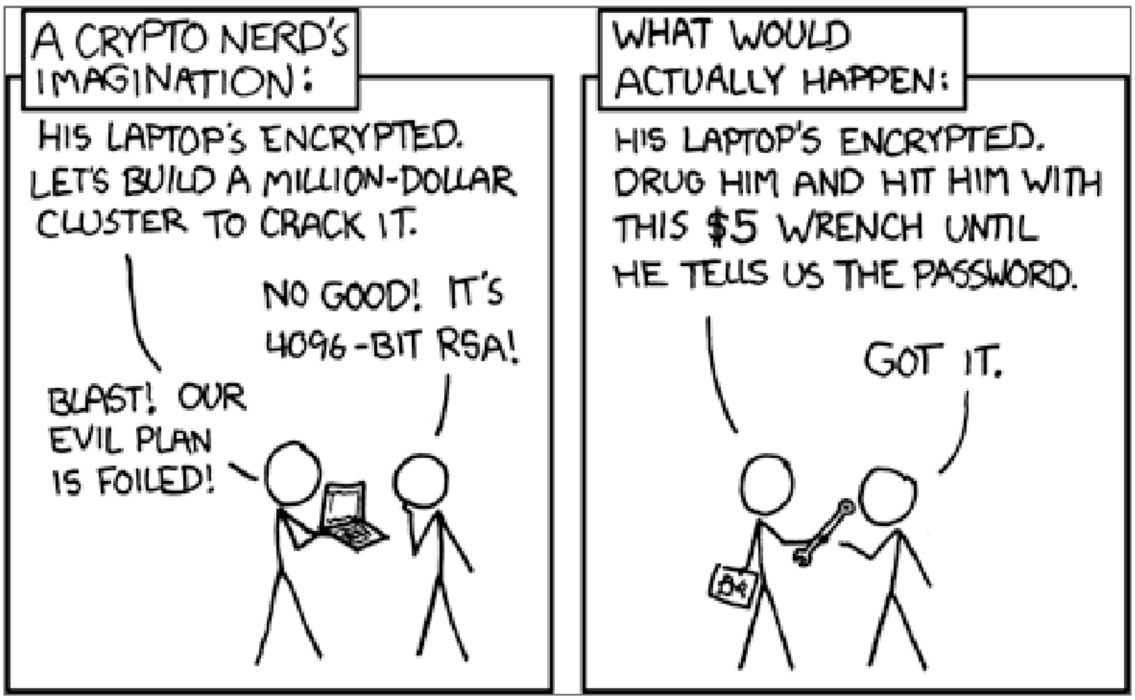
Enforce policy carefully

- Fail-safe defaults
- Complete mediation
- Separation of privilege

> ## Limit blast radius
>
> - Least privilege
> - Least common mechanism
> - Acceptability

Encouraging good security

- **Weakest links will break first**
- Breaking a cryptographic hash is really hard
- Breaking “password” (i.e. the literal word as a password) is really easy



System Calls - Not all are equal

- `exit()`
- `getpid()`
- `getppid()`
- `time()`
- `wait()`
- `fork()`
- `mmap()`
- `mprotect()`
- `socket()`
- `recv()`

Summary

- **Security is essential to any system**
- **Virtualization and sharing are useful, but they create policy decisions that the OS must enforce**
- **The security goals are confidentiality, integrity, and availability**
- **Good security comes from simple mechanisms, careful checks, and designs people can actually use**
- **In practice: focus on what works, and make sure it still works when things go wrong**

Speaker notes